MALWARE SCANNER PROTOTYPE

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CERTIFICATE

This is to certify that the Subject Specific Project entitled "**Malware Scanner Prototype**" has been carried out by 1. **Subham Khandual (2301298076)**, 2. **Purusottam Nanda (2301298196)** & 3. **Subham Kumar Panda (2301298246)** and completed under my guidance and the project meets the academic requirement of the project.

Signature of the guide

TABLES OF CONTENT

Sr. No	Contents	Page No
1	Abstract	1
2	Introduction	2
3	Literature Survey	3
4	Problem Statement	4
5	Method	5
6	Source code	6
7	Result/Outcome	7
8	Conclusion	8
9	References	9

ABSTRACT

In today's digital world, cybersecurity threats such as malware, ransomware, and spyware pose a significant risk to personal and organizational data. To mitigate these threats, a **Malware Scanner Prototype** is developed as a preventive security tool that detects, analyzes, and removes malicious files from computer systems. The proposed system employs signature-based and heuristic analysis techniques to identify known and unknown malware efficiently.

Signature-based detection matches files against a database of known malware signatures, while heuristic analysis evaluates suspicious behaviors and file structures to catch new or modified threats. The prototype provides a user-friendly interface for scanning specific files, folders, or entire systems and generates detailed reports of detected threats.

By integrating automation and real-time monitoring, the malware scanner enhances system security and helps users maintain a safe computing environment. This project demonstrates the core functionality of antivirus mechanisms and serves as a foundation for developing advanced, AI-based malware detection systems in the future.

INTRODUCTION

With the rapid advancement of technology and the increasing use of the internet, cybersecurity has become a major concern for individuals, organizations, and governments worldwide. Malware, a collective term for malicious software such as viruses, worms, Trojans, spyware, and ransomware, poses a serious threat to data integrity, system performance, and user privacy. As cyberattacks become more sophisticated, the need for effective detection and prevention mechanisms has grown significantly.

A **Malware Scanner Prototype** is designed to address these challenges by providing a reliable and efficient way to identify and remove harmful software from computer systems. The scanner works by analyzing files and applications for suspicious patterns, code behavior, and known malware signatures. It can detect both existing and newly emerging threats through a combination of signature-based detection and heuristic analysis.

This prototype aims to demonstrate the fundamental working principles of antivirus software, offering real-time scanning, threat detection, and reporting capabilities. By creating a user-friendly interface and automated scanning process, the system enables users to maintain their devices securely and efficiently. Ultimately, the project contributes to the growing field of cybersecurity by offering a foundation for more advanced, intelligent malware detection systems that can adapt to evolving cyber threats.

LITERATURE SURVEY

Malware detection has been a critical research area in the field of cybersecurity due to the increasing frequency of cyberattacks and system vulnerabilities. Early studies primarily focused on **signature-based detection**, which identifies malicious software by comparing files against a database of known virus signatures. To overcome this limitation, researchers introduced **heuristic-based detection**, which analyses program behavior and structural patterns to identify suspicious files. Although heuristic methods improve detection of unknown malware, they sometimes produce false positives, classifying safe files as threats.

Further advancements led to the development of **behavioural and hybrid detection techniques**. Studies also explored **static and dynamic analysis** methods. Static analysis inspects files without execution, making it safe but limited against encrypted or obfuscated malware. Dynamic analysis executes files in a sandbox environment to monitor their actions but requires more system resources.

In recent years, researchers have applied **machine learning and artificial intelligence (AI)** techniques to malware detection. ML models can automatically classify malware by learning from large datasets of malicious and benign samples, enabling the detection of zero-day and polymorphic malware. Overall, literature suggests that **hybrid and AI-based malware scanners** provide a more robust, adaptive, and efficient solution for detecting both known and unknown threats, forming the basis for modern malware scanner prototypes.

PROBLEM STATEMENT

1. he rapid increase in internet usage has led to a rise in malware attacks such as viruses, worms, Trojans, spyware, and ransomware.
2. Malware can cause serious damage by stealing data, corrupting files, slowing system performance, and compromising user privacy.
3. Most existing antivirus systems mainly depend on **signature-based detection**, which is effective only for known threats.
4. New and modified malware (zero-day attacks) often go undetected by traditional signature-based scanners.
5. Many commercial antivirus solutions are **expensive** and **resource-intensive**, making them less suitable for educational or lightweight applications.
6. There is a lack of affordable and customizable tools that combine **signature-based** and **heuristic-based** detection techniques for improved accuracy.
7. Hence, there is a need to develop a **Malware Scanner Prototype** that can:
    - Detect both known and unknown malware effectively.
    - Analyze suspicious files using multiple detection methods.
    - Provide real-time scanning and detailed reporting.
    - Offer a simple, user-friendly interface for ease of use.
8. The goal is to create a **cost-effective, efficient, and reliable malware detection system** that enhances computer security and helps users safeguard their data.

METHOD

Requirement Analysis:

- Identify the system requirements such as programming language, scanning algorithms, database for signatures, and user interface design.
- Study existing malware detection techniques to select appropriate methods for the prototype.

Data Collection:

- Collect datasets of known malware signatures and sample files for testing.
- Prepare a database containing hash values or pattern signatures of malicious files.

System Design:

- Design the architecture of the malware scanner including input module, scanning engine, detection module, and report generator.
- Develop a user-friendly graphical interface for easy interaction.

Implementation:

- Implement **signature-based detection** to identify known malware using stored signatures.
- Implement **heuristic analysis** to detect unknown or suspicious files by analyzing code structure and behavior.
- Integrate **static and dynamic analysis** to improve detection accuracy.

Scanning Process:

- The user selects files, folders, or drives for scanning.
- The scanner compares file data with the signature database and applies heuristic rules for suspicious patterns.
- Detected malware is flagged, quarantined, or deleted as per user choice.

Testing and Evaluation:

- Test the prototype with various malware samples and clean files.
- Evaluate performance in terms of detection accuracy, speed, and false positives.

Result and Improvement:

- Analyze test results to identify strengths and weaknesses.
- Update the signature database and refine heuristic rules for better detection.

SOURCE CODE

PHISHGUARD
- __pycache__
- instance
- screenshots
- static
- templates
- venv
- .gitattributes
- .gitignore
- app.py
- extensions.py
- LICENSE
- models.py
- phishing_detector.py
- Procfile
- README.md
- requirements.txt
- routes.py

app.py > ...

```python
import os
import logging
import json
from flask import Flask
from werkzeug.middleware.proxy_fix import ProxyFix
from extensions import db  # single shared db instance

logging.basicConfig(level=logging.DEBUG)

# Create Flask app
app = Flask(__name__)
app.secret_key = os.environ.get("SESSION_SECRET", "dev-secret-key-change-in-production")
app.wsgi_app = ProxyFix(app.wsgi_app, x_proto=1, x_host=1)

# Database configuration
app.config["SQLALCHEMY_DATABASE_URI"] = os.environ.get(
    "DATABASE_URL", "sqlite:///phishing_detector.db"
)
app.config["SQLALCHEMY_ENGINE_OPTIONS"] = {
    "pool_recycle": 300,
    "pool_pre_ping": True,
}

# Bind db to app before importing models or routes
db.init_app(app)

```

OUTLINE

TIMELINE

PHISHGUARD
- __pycache__
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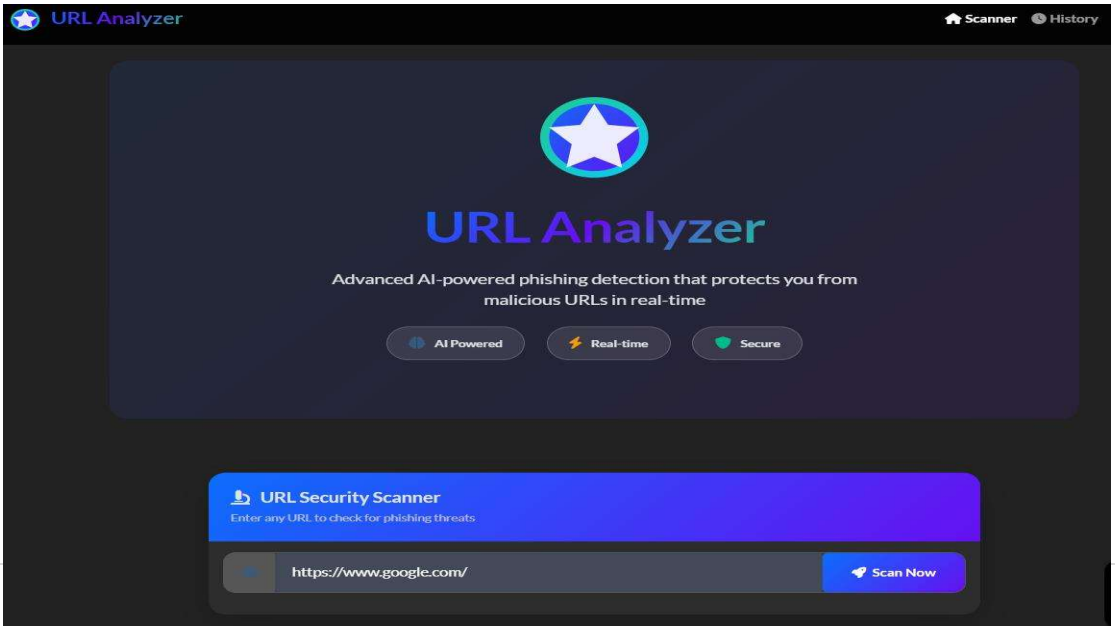
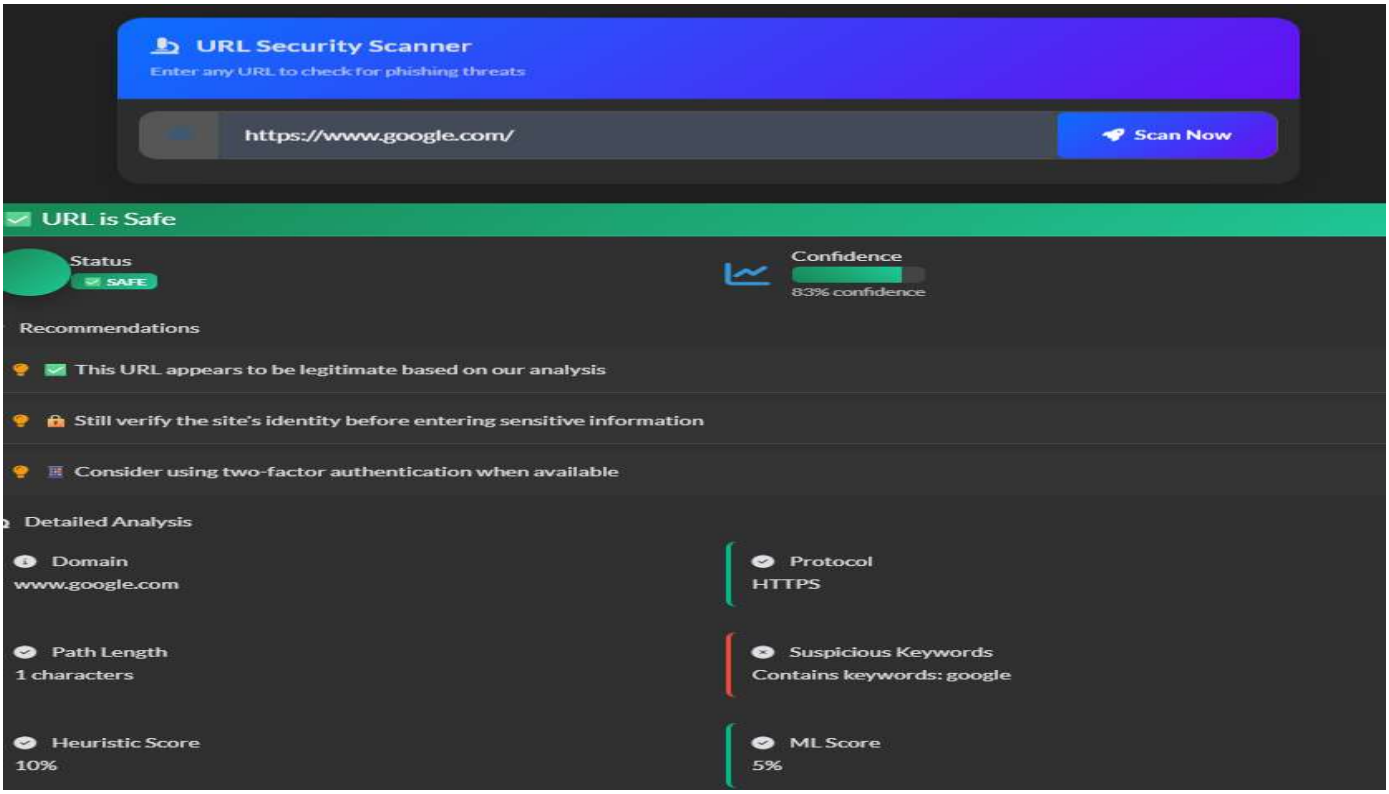
app.py > ...

```python
# Create tables after binding db
with app.app_context():
    # Import models after db is initialized
    from models import URLAnalysis
    db.create_all()

# Import routes after db + models are ready
from routes import main_bp

app.register_blueprint(main_bp)

# Add custom Jinja2 filter for from_json
@app.template_filter('from_json')
def from_json_filter(value):
    if value:
        return json.loads(value)
    return {}

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=True)

```

RESULT/OUTCOME





CONCLUSION

The **Malware Scanner Prototype** successfully demonstrates the fundamental working of a malware detection system by integrating signature-based and heuristic analysis techniques. It provides an efficient way to identify and remove both known and unknown malicious files from a computer system. Through the use of static and dynamic analysis methods, the prototype enhances the accuracy of threat detection while maintaining a user-friendly and automated interface.

This project highlights the importance of multi-layered security approaches in combating evolving cyber threats. The results show that combining different detection techniques reduces false positives and improves overall system protection. Although the prototype is a basic model, it serves as a foundation for developing advanced malware scanners that can incorporate artificial intelligence and machine learning for more accurate and adaptive detection in the future.

Overall, the malware scanner prototype contributes to the growing field of cybersecurity by providing a cost-effective, flexible, and reliable solution for ensuring data safety and system integrity.

REFERENCE

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